Andrew Boldi

Solving hard science problems with machine learning

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Education

UC Berkeley *Expected May 2027*

BS in Chemical Biology & BA in Computer Science

Fall '26:	Deep Neural Networks (CS 182), Computer Vision (CS 180), Physical Chemistry (Statistical Mechanics), Molecular Biology Lab
Chemistry:	Computational Chemistry, Physical Chemistry (Quantum Mechanics), Honors Organic Chemistry I/II, Honors General Chemistry I/II
Computer Science:	ML (CS 189), AI (CS 188), Data Structures & Algorithms (CS 61B), Computer Architecture (CS 61C), Discrete Math & Probability Theory (CS 70), Structure & Interpretation of Computer Programs, Designing Information Devices & Systems
Sciences:	Macromolecular Biology, Chemical Biology (Biochemistry), Biological Inorganic Chemistry, Human Physiology, Mechanics, Electricity & Magnetism, Chemical Engineering Process Analysis, Mathematics & Statistics for Chemical Engineering

Publications

A. Boldi[†], A. K. Gupta[†], Y. Tang, M. Hamadani, K. Ameri, Y. Yeghiazarians. "Fragment-Centric Explainability of Chemical Foundation Models with Attention-Aware Layer-Wise Relevance Propagation." *Under review*, 2026. [†]Co-first authors.

Work Experience

AI Intern *South San Francisco, CA*

Soley Therapeutics: Molecular Machine Learning under direction of Dr. Ankur Gupta *May 2025 – Present*

- Full-time summer 2025 and 2026 intern; received return offer and continued projects part-time during the academic year
- **Co-first author** (manuscript under review) and **inventor** (filed patent application) on a fragment-centric explainability framework for chemical foundation models; co-led its development, applied it to Ames mutagenicity, hERG inhibition, and lipophilicity models, and demonstrated usefulness in internal lead optimization
- Leading development of explainability methods for computer vision models on cell images
- Finetuned hundreds of models on public and proprietary data to predict molecular target candidate profiles (TCPs); built an automated framework for configuration generation, finetuning, and cross-fold validation of several foundation models across 22 datasets (TDC ADMET group)
- Developing generative chemistry models for de novo molecule generation and lead optimization of therapeutics
- Implemented and benchmarked several architecture ideas for molecular property prediction foundation models; refactored the codebase of an existing foundation model
- Automated CDD (Collaborative Drug Discovery) protocol creation with Selenium + Python, completing a task projected to take 100+ manual hours in 5–6 hours
- Integrated the above projects into the product with the UI/Infra team
- Tools used: PyTorch, PyTorch Lightning, Hydra/OmegaConf, CDD Vault, AWS (EC2, S3, CodeBuild)

Research Experience

Summer Research Intern *Pasadena, CA*

Caltech: Professor Brian M. Stoltz Lab Group under direction of Dr. Scott Virgil *Jun 2023 – Aug 2023*

- Completed 16 consecutive reaction steps toward an intermediate in the total synthesis of the diterpenoid scabrolide B
- Developed software application for cataloging catalysts & ligands (StremLigand)
- Performed computational modeling (ORCA) for determining selectivity in a late-stage stereoinversion
- Used lab techniques including HPLC, LC/MS, GC/MS, LC-TOF, NMR, gCOSY 2D NMR, TLC, extraction, column chromatography, rotary evaporator, distillation, dry box

Summer Research Intern *Pasadena, CA*
Caltech: Professor Brian M. Stoltz Lab Group under direction of Dr. Scott Virgil *Jun 2022 – Aug 2022*

- Completed the 10-step synthesis of a fragment in the total synthesis of korormicin
- Completed the 8-step synthesis of a fragment in the total synthesis of ineleganolide
- Conducted computational modeling of key intermediates to identify optimal synthesis routes using ORCA
- Taught myself CAD (FreeCAD) overnight, then measured, modeled, and 3D printed a replacement gear for a broken HPLC autosampler motor the same day; refined it to an exact fit, avoiding a $250 part with a two-week lead time, and printed spares now also used in a Caltech neurobiology lab

Assistant Researcher *Remote*
Caltech: Center for Catalysis and Chemical Synthesis under direction of Dr. Scott Virgil *Jul 2020 – May 2022*

- Developed software program for remote automation of robotic autosamplers (Python, VBA, M68k assembly language)
- Reverse-engineered M68k assembly code to write more efficient functions for use in lab
- Used program to screen rare earth metal (lanthanide)-ligand catalysts for an asymmetric reduction

Research Summaries & Proposals

Computational Total Synthesis Project Proposal *Mar 2024*

Deep Reinforcement Learning for Total Synthesis

Synthetic Efforts Towards the Total Synthesis of Scabrolide B *Aug 2023*

2023 Summer Research

Synthesis of Korormicin and Ineleganolide Fragments *Aug 2022*

2022 Summer Research

Projects

Molecular Optical Rotation Prediction with Machine Learning — *Trennen* *2022*

- Self-directed project: distilled 8M+ molecules down to ∼27,000 optically active molecules using PubChem's API, RDKit, and Open Babel
- Built a T-Net (point-cloud) classifier on 3D molecular structures to predict rotation direction
- Explored computing polarizability tensors with ORCA at the B3LYP/def2-SVP level of theory (DFT)
- Tools used: TensorFlow, PubChem REST API, Open Babel, RDKit, ORCA

Scenic Navigation Algorithm with Deep Learning — *Pathfinder* *2022*

- Developed navigation algorithm (modified A*) to find scenic routes in San Mateo County
- Developed web-scraping program with Selenium to automatically extract 250K+ Google Street View images from San Mateo County
- Used a pretrained semantic segmentation model for inference on 250K+ images with Google TPUs
- Accounted for curvature of the earth with mathematical expressions for better heuristic estimation
- Processed road connections and intersections with United States Census Bureau's TIGER/Line Shapefiles
- Tools used: Python, BDD100K Dataset, Google TPUs

Pacman with Bitcoin Miner (Top Project in Berkeley Course) — *Video Demo* *Spring 2024*

- Built a Pacman game with a built-in Bitcoin miner with a project partner in 3000+ lines of Java code
- Selected as one of the top 15 projects in a class of ∼1400 students for a data structures & algorithms course from UC Berkeley (CS 61B, Spring 2024)

Math Competition Training Website — *AMC Trainer* *2021 – Present*

- Co-developed interactive website and backend with Ethan Boldi to help students prepare for math competitions (grades 8/10/12 and invitational AIME)
- Actively used by 20K+ users in 50+ countries
- Tools used: JavaScript, HTML, CSS, Cloudflare Workers, Cloudflare KV Database, Node.js

Representation of the Fibonacci and Prime Numbers as Music — *Fibprimes* *2021*

- Built a program in C to automatically write MIDI files with the pattern of Fibonacci and prime numbers

- Recognized on Y Combinator's Hacker News and earned 26 stars on GitHub
- Tools used: C, MIDI file format

Skills

Programming:	Python, Java, C/C++, Rust, Svelte, JS/CSS/HTML, M68k Assembly, Swift, LaTeX
ML & Tools:	PyTorch, PyTorch Lightning, TensorFlow, Hydra/OmegaConf, Selenium, AWS (EC2, S3, CodeBuild), Git/GitHub, Slurm, Linux
Chemistry:	Multistep organic synthesis, laboratory characterization (HPLC, LC/MS, NMR), computational modeling (ORCA, DFT), RDKit, AutoDock Vina
Languages:	English, Armenian, Spanish

Additional Experience

Physician Shadowing at Kaiser Permanente *Summer 2026*

- Shadowed Kaiser physicians and engineers in general surgery, magnetoencephalography (MEG), hospital medicine, and palliative medicine

Code Mind Berkeley: Co-founder *Nov 2023 – Jan 2024*

- Interdisciplinary research for high school students with UC Berkeley professors
- Built website with Svelte + SvelteKit + TypeScript + SCSS + GSAP
- Developed research project proposals at the intersection of machine learning, analytical chemistry, physics, biochemistry, marine science, and medicine
- Developed web-scraping pipeline for generating leads in Python + Selenium, later rewritten in Rust
- Secured $250K in client commitments (75 clients); wound down the company to focus on full-time studies

Y Combinator AI Startup School *Jun 2025*

- Selected as one of 2000 top CS undergrads, master's, and PhD candidates in AI to attend the conference, with speakers including Elon Musk, Satya Nadella, Sam Altman, Andrej Karpathy, Andrew Ng, and more
- Further invited as one of ∼20 students to have dinner with YC general partner Aaron Epstein

Private Tutor *2018 – Present*

- Teaching 7 students math and piano weekly
- Taught 25+ students in the past in subjects including linear algebra and chemistry

STEMPower: Workshop Facilitator & Program Director *Jul 2021 – Jun 2023*

- Led hands-on workshops through Zoom to promote scientific interest for grades 3–6
- Subjects taught include math, chemistry, biology, and physics

YAPA Kids: Math Tutoring Teacher & Program Director *May 2020 – Aug 2021*

- Led and taught interactive math classes for enhanced understanding of school topics
- Math Co-Program director for over 460 students from May 2021 to August 2021

Special Interests

Rubik's Cube Puzzles

- Solved 3x3 in 9.42 seconds & have solved Rubik's cube blindfolded
- Skilled at rapidly solving 2x2, 3x3, 4x4, 5x5, 6x6, 7x7, Megaminx (12-sided), and Square-1 cubes

Music

Classical pianist

@AndrewBoldiPiano

Highlights

- Began classical piano study at age 7½; 30+ recognitions and 20+ public performances in classical piano
- Multiple 1st prizes at competitions including US Open Music, Armenian Allied Arts Association, Crestmont Conservatory Piano, and Satori Young Pianist; twice awarded "Most Promising Performer" at the Armenian Allied Arts Competition
- Performed at Carnegie Hall (Weill Recital Hall) and Merkin Concert Hall in New York City as a prize winner of the Armenian Youth Talent Competition
- Helped raise $285K, $255K, and $65K for Armenian churches in San Francisco and Pasadena as the only solo pianist at their banquets; performer at local Armenian community events and senior homes around the Bay Area
- Completed the final advanced level of the MTAC Certificate of Merit Program in Piano at age 12

Competitions & Awards

United States Open Music Competition, *Oakland, CA*

- 1st Place, Showcase Piano Solo Intermediate (Ages 13–15); recognized for one of the best performances at the Winner's Concert (interviewed live on stage) — *Feb 17, 2020*
  Beethoven, Sonata No. 23, F min, Op. 57 (1st mov't) 'Appassionata'; Babajanian, Capriccio
- 1st Place, Showcase Piano Solo Junior (Ages 10–12); recognized for one of the best performances at the Winner's Concert — *Feb 23, 2017*
  Bach, Prelude & Fugue G min BWV 861; Babajanian, Vagharshapat Dance
- 2nd Place, Showcase Piano Solo Intermediate (Ages 13–15) — *2021*
  Liszt, La Campanella, Grandes études de Paganini S. 141 No. 3 in G sharp minor; Beethoven, Sonata No. 17, D min, Op. 31, No. 2 (3rd mov't) 'Tempest'
- 2nd Place, Treasury of Baroque Composers Advanced (Ages 7–19) — *Feb 20, 2020*
  Bach, Prelude & Fugue D minor BWV 875
- 2nd Place, Treasury of Romantic Composers (Chopin Etudes) Advanced (Ages 7–19) — *Feb 19, 2019*
  Chopin, Etude C# minor, Op. 25, No. 7
- 2nd Place, Treasury of Romantic Composers (Chopin Etudes) Advanced (Ages 7–19) — *Feb 23, 2017*
  Chopin, Etude F minor, Op. 10, No. 9
- 3rd Place, Treasury of Romantic Composers (Rachmaninoff Etudes) Advanced (Ages 7–19) — *Feb 21, 2020*
  Rachmaninoff, Etude-Tableaux, Eb Maj, Op. 33, #4 (o7)
- 3rd Place, Showcase Piano Solo Intermediate (Ages 13–15) — *Feb 21, 2019*
  Beethoven, Sonata No. 17, D min, Op. 31, No. 2 (1st mov't) 'Tempest'; Debussy, Prelude A minor, Pour le piano
- 4th Place, Showcase Sight-Reading Solo Intermediate–Senior — *May 15, 2021*
- 4th Place, Treasury of Romantic Composers (Chopin Etudes) Advanced (Ages 7–19) — *Feb 2018*
- 4th Place, Treasury of Romantic Composers (Chopin Etudes) Advanced (Ages 7–19) — *Feb 16, 2016*
- 4th Place, Showcase Piano Solo Junior (Ages 10–12) — *Feb 16, 2016*

Armenian Allied Arts Competition, *Pasadena, CA*

- 1st Place and Hovsep & Elise Ignatius Award for "Most Promising Performer" as Judges' Favorite Performance, Juniors (Ages 12–16), 78th Annual Competition — *Jun 2018*
- 1st Place and Hovsep & Elise Ignatius Award for "Most Promising Performer" as Judges' Favorite Performance, Juniors (Ages 12–16), 77th Annual Competition — *Apr 2017*

Satori Young Pianist Contest, *Arcadia, CA*
5th Satori Young Pianist Contest, 2020 Satori Summer Music Festival

- 1st Place, Prodigy Category III (Ages 13–15) — *Aug 1, 2020*
  Beethoven, Sonata No. 23 in F minor, Op. 57, 'Appassionata', I. Allegro Assai
- 1st Place, Open Category III (Ages 13–15) — *Aug 1, 2020*
  Liszt, La Campanella, Grandes études de Paganini S. 141 No. 3 in G sharp minor

Crestmont Conservatory Piano Competition, *San Mateo, CA*

- 1st Place, Category II (Ages 11–13) *Jan 31, 2016*
- 1st Place, Category I (Ages 7–10) *Feb 9, 2015*
- Honorable Mention, Category III (Ages 14–18) *Jan 29, 2017*

San Carlos Has Talent, *San Carlos, CA*

- 1st Place, 6th Annual San Carlos Has Talent (General Music Talent Competition) *Nov 13, 2016*

San Mateo County Fair Classical Piano Youth Competition, *San Mateo, CA*

- 2nd Place, Ages 11–12 Division *Jun 13, 2017*
- 2nd Place, Ages 11–12 Division *Jun 14, 2016*
- 2nd Place, Ages 9–10 Division *Jun 7, 2015*

United States International Music Competition, *Stanford, CA*

- 3rd Place, Special Theme Award – J. S. Bach Piano C (Ages 12–13) *Aug 13, 2017*
- Honorable Mention, Group C (Ages 12–14) *Aug 17, 2019*
  Beethoven, Sonata No. 17, D min, Op. 31, No. 2 (1st mov't) 'Tempest'; Bagdasarian, Prelude No. 6, B min
- Honorable Mention, Group B (Ages 10–11) *Aug 12, 2016*
- Finalist, Young Artist Award (Ages 15–18) *Sep 2020*
- Finalist, Young Artist Award (Ages 12–13) *Aug 2018*

Burlingame Music Club Music Competition

- Roz Koo Memorial Award *2021*
  Beethoven, Sonata No. 23 in F minor, Op. 57, 'Appassionata', I. Allegro Assai; Liszt, La Campanella, Grandes études de Paganini S. 141 No. 3 in G sharp minor, Allegretto; Babajanian, Capriccio in D flat Major, Allegro Moderato

San Jose International Piano Competition, *San Jose, CA*
Formerly the International Russian Music Piano Competition

- Finalist, Junior Category (Ages 10–13), 17th competition *Jun 12, 2016*
- Finalist, Young Musicians Category (Ages 7–9), 16th Annual competition *Jun 6, 2015*

Los Angeles Young Musicians International Competition, *Los Angeles, CA*

- Finalist, Category C (Ages 10–11), 4th Annual competition *Aug 4, 2016*
- Finalist, Category C (Ages 10–11), 4th Annual competition *Jul 23, 2015*

Selected Performances

Armenian Youth Talent Competition Winner's Concerts, *New York, NY*
Performed as a prize winner; concerts benefiting Direct Help for Armenia

- **Weill Recital Hall, Carnegie Hall** *Oct 20, 2018*
- **Merkin Concert Hall**; performed and interviewed for an Armenian television program *Sep 17, 2017*

Junior Bach Festival

- Virtual performance, due to the pandemic, in lieu of a live performance at the 67th Annual Festival, Christ Episcopal Church, Portola Valley, CA *Mar 21, 2020*
- Selected to perform at the 64th Annual Festival, Berkeley Piano Club, Berkeley, CA; performed several Preludes & Fugues *Mar 24, 2017*

United States Open Music Competition Winner's Concerts

- Performed as First Place Gold Medalist *Feb 17, 2020*
- Performed as First Place Gold Medalist *Feb 25, 2017*

Music Teachers Association of California (MTAC)

- San Mateo County MTAC Branch Honors Recital, Foster City, CA *May 5, 2019*
- Annual State MTAC Convention *Jul 30, 2018*
- San Mateo County MTAC Branch Honors Recital, Foster City, CA *Mar 2018*

Original Arrangement & Performance — *Video*

- *O, The Deep, Deep, Love of Jesus*: original arrangement based on Chopin Piano Concerto No. 1 — *Oct 23, 2023*

Burlingame Music Club

- Virtual performance — *2021*
- Performance at The Stratford – Senior Living, San Mateo, CA — *Mar 22, 2019*

Yoko Okada Piano Recital, *San Mateo Library, San Mateo, CA*

- Guest soloist — *Feb 2, 2020*

Benefit & Community Performances

St. Gregory the Illuminator Armenian Church of San Francisco, *San Francisco, CA*
Armenian church targeted in a 2020 hate-crime arson attack

- Only solo pianist at the 56th Anniversary Celebration (∼350 guests); helped raise $255K performing works by Babajanian and Liszt — *Oct 16, 2022*
- Only solo pianist at the 55th Anniversary Celebration (∼350 guests); helped raise $285K performing works by Babajanian — *Nov 7, 2021*

Armenian Brotherhood Bible Church, *Pasadena, CA*

- Only solo pianist, 50th Church Anniversary Banquet (∼400 guests); helped raise $65K — *Nov 5, 2022*
- Only solo pianist, Thanksgiving Banquet (∼400 guests) — *Nov 17, 2018*
- Only solo pianist, Thanksgiving Banquet (∼400 guests) — *Nov 14, 2015*
- Only solo pianist, Thanksgiving Banquet (∼400 guests) — *Nov 22, 2014*

Benefit Banquet for Armenians in Syria & Lebanon, *Fresno, CA* — *Video*
Sponsors: Pilgrim Armenian Congregational & First Armenian Presbyterian Churches

- Performed *O, The Deep, Deep, Love of Jesus* (arr. Vocalise, Rachmaninoff), *Capriccio* (Babajanian), and *La Campanella* (Liszt) — *Mar 11, 2023*

Hope House (a foster home for Mexican children), *Sunnyvale, CA*

- At age 11, the only minor soloist (among mostly Stanford students) invited to play in a benefit concert, which raised ∼$10K; performed Rachmaninoff, *Prelude C# minor* — *Oct 23, 2015*

Hamazkayin Armenian Educational and Cultural Society, *San Francisco, CA*

- Virtual concert soloist for *Armenian Relief Cross for Lebanon* — *Oct 22–24, 2021*
- Only solo pianist at the 2018 Hamazkayin 60th Anniversary Celebration (∼300 guests), performing a program of works by Bagdasarian and Babajanian — *Sep 22, 2018*

Calvary Armenian Congregational Church, *San Francisco, CA*

- Only solo pianist at banquet (∼250 guests), performing works by Bagdasarian and Chopin — *May 12, 2018*
- Only solo pianist at banquet (∼250 guests), performing works by Babajanian and Chopin — *Jun 10, 2017*

Nursing Homes

- San Carlos Elms Nursing Home, San Carlos, CA — *Jan 25, 2020*
- Carlmont Gardens Nursing Home, Belmont, CA — *Dec 15, 2018*

Ensembles & Leadership

Conductor, Carlmont Pep Band — *Sep – Oct 2022*

- Led the 200+ person band at three home football games and two school events

Pianist, Carlmont High School Jazz Ensemble — *2019 – 2023*

- Sole pianist in the ensemble for 4 years
- Winter Concerts, Jazz Ensemble and Jazz Combo — *2019 – 2021*
- Spring Concerts, Jazz Ensemble and Jazz Combo — *2020 – 2022*
- CMEA Jazz Festival — *2019*

Accompanist, Carlmont High School Chamber Music Night — *2022*

- Euphonium duet and violin duet

Training & Certification

MTAC Certificate of Merit Program in Piano

- Completed the final advanced level at age 12, with the highest performance scores for all repertoire and a perfect score on the music theory exam

Dr. Michael Schmitz, Private Piano Teacher, Classical Piano — *2014 – 2021*
Professor Emeritus, Notre Dame de Namur, Belmont, CA

Brian Switzer, Director, Carlmont High School Jazz Ensemble — *2019 – 2023*
Carlmont High School, Belmont, CA

Yoko Okada, Private Piano Teacher — *2013 – 2014*
San Carlos, CA

Recordings & Repertoire

Andrew Boldi Piano — *YouTube Channel*

- Repertoire featuring Bach, Beethoven, Chopin, and Armenian composers Babajanian, Bagdasarian, and Khachaturian
- Live performance recording: Babajanian, *Capriccio*; Liszt, *La Campanella* (earlier recordings below) — *Nov 5, 2022*

Select Repertoire *(titles link to recordings on YouTube)*

Bach, Johann Sebastian:	English Suite No. 3, G min, BWV 808; Prelude & Fugue, C min, BWV 847; Prelude & Fugue, G Maj, BWV 860; Prelude & Fugue, G min, BWV 861; Prelude & Fugue, D min, BWV 875; Prelude & Fugue, F min, BWV 881; Prelude, C min, BWV 999
Beethoven, Ludwig van:	Sonata No. 11, Bb Maj, Op. 22 (1st mov't); Sonata No. 11, Bb Maj, Op. 22 (2nd mov't); Sonata No. 17, D min, Op. 31, No. 2 (1st mov't) 'Tempest'; Sonata No. 17, D min, Op. 31, No. 2 (3rd mov't) 'Tempest'; Sonata No. 19, G min, Op. 49, No. 1 (1st mov't); Sonata No. 23, F min, Op. 57 (1st mov't) 'Appassionata'
Babajanian, Arno:	Capriccio (2020); Elegy; Vagharshapat Dance
Bagdasarian, Eduard:	Prelude No. 6, B min
Chopin, Frederic:	Etude, C# min, Op. 10, No. 4 'Torrent'; Etude, F min, Op. 10, No. 9; Etude, C min, Op. 10, No. 12 'Revolutionary'; Etude, F min, Op. 25, No. 2 'The Bees'; Etude, C# min, Op. 25, No. 7 'Cello'; Etude, Gb Major, Op. 25, No. 9 'Butterfly'; Etude, C min, Op. 25, No. 12 'Ocean'
Debussy, Claude:	Prélude, A min, Pour le piano
Hicken, David:	Carol of the Bells
Khachaturian, Aram:	Toccata, Eb min
Liszt, Franz:	La Campanella (2021)
Rachmaninoff, Sergei:	Etude-Tableaux, Eb Maj, Op. 33, #4 (o7); Prelude, C# min, Op. 3, No. 2